Miyawaki Native Afforestation and Campus Carbon Sequestration Report

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MIYAWAKI NATIVE AFFORESTATION AND CAMPUS CARBON SEQUESTRATION REPORT

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Project Identifier: INIT_BIODIVERSITY_ZONE (Miyawaki Native Afforestation Project)

Executing Unit: Campus Nature Club and Environmental Civil Cell.

Advisory Support: Social Forestry Division, Government of Maharashtra.

Location: Vivekananda Education Society's Institute of Technology (VESIT Green Campus), Chembur, Mumbai.

1. BOTANICAL PROJECT GENESIS AND METHODOLOGY

In June 2023, Vivekananda Education Society's Institute of Technology (VESIT Green Campus) dedicated a 600 m2 peripheral parcel of land located immediately south of the Auditorium building (BLD_AUD) to establish a high-density native micro-forest. The project deployed the internationally recognized Miyawaki afforestation methodology pioneered by Japanese botanist Dr. Akira Miyawaki. This ecological technique mimics natural forest succession by planting indigenous multi-canopy trees at high density (3 to 4 saplings per square meter), stimulating rapid upward growth through competitive sunlight seeking and accelerated root mycorrhizal networks.

Soil preparation involved excavating topsoil to a depth of 1.0 meter, incorporating organic perforated amendments: 4,000 kg of composted bioslurry produced on-campus by the Canteen anaerobic digester (INIT_BIOGAS_CANTEEN), coco-peat for moisture retention, rice husk for aeration, and indigenous micro-organisms. A total of 1,200 saplings across 32 indigenous tree and shrub species native to the Northern Western Ghats and Konkan coastal ecosystem were planted in July 2023.

2. SPECIES COMPOSITION AND MULTI-LAYER CANOPY STRATIFICATION

The botanical inventory was organized across four structural canopy tiers:

- Canopy Emergent Trees (Overstory, 15-20m potential height): Ficus religiosa (Peepal), Ficus benghalensis (Banyan), Azadirachta indica (Neem), Syzygium cumini (Jamun), and Mangifera indica (Native Mango).

- Sub-Canopy Trees (10-15m): Pongamia pinnata (Karanj), Bauhinia variegata (Kanchan), Cassia fistula (Amaltas), and Alstonia scholaris (Saptaparni).

- Understory Small Trees & Large Shrubs (5-10m): Morinda citrifolia (Noni), Phyllanthus emblica (Amla), and Bambusa multiplex (Hedge Bamboo).

- Ground Cover & Medicinal Shrubs: Adhatoda vasica (Adulsa), Ocimum sanctum (Tulsi), and Rauvolfia serpentina (Sarpagandha).

3. ECOLOGICAL SURVIVAL METRICS AND MICROCLIMATIC IMPACTS

At the comprehensive 12-month ecological audit completed in June 2024:

- Survival Rate: Recorded an exceptional 94.2% survival rate (1,130 surviving trees out of 1,200 planted), attributed to drip irrigation utilizing treated canteen greywater and rainwater cistern supplies from INIT_RWH_SYSTEM.

- Canopy Growth: Average canopy height surpassed 3.8 meters, with pioneer species (Karanj and Bamboo) exceeding 5.2 meters. Canopy foliage density has created a closed-canopy micro-ecosystem.

- Biodiversity Index: Avian and pollinator surveys conducted by student volunteers documented 24 resident and migratory bird species (including oriental magpie robins, sunbirds, and coppersmith barbets) alongside 18 butterfly species.

- Microclimate Cooling: Thermal infrared sensor arrays deployed under INIT_IOT_ENERGY_MGT measured a localized ambient temperature depression of 2.3°C within the forest interior compared to adjacent paved asphalt parking surfaces, significantly mitigating the local urban heat island phenomenon.

4. ALLOMETRIC BIOMASS AND CARBON SEQUESTRATION MODELING

Carbon sequestration was quantified using standard non-destructive allometric biomass equations developed by the Forest Survey of India (FSI) for tropical broadleaf species:

Biomass (kg) = 0.0509 * (Diameter at Breast Height in cm)^2 * Height (m) * Wood Density

- Current Measured Sequestration (Year 1): 14,000 kg CO2e/year (14.0 metric tonnes CO2e).

- Projected Maturity Sequestration (Year 5 - 2028): 26,500 kg CO2e/year.

- Projected Climax Sequestration (Year 10 - 2033): 38,000 kg CO2e/year.

The project contributes tangible biotic carbon removal advancing target TRG_NET_ZERO, directly supporting United Nations SDG 13 (Climate Action) and SDG 15 (Life on Land). Future plans incorporate expanding the afforestation corridor by an additional 400 m2 along the eastern campus perimeter.

INSTITUTIONAL AUDIT & COMPLIANCE VERIFICATION

*This statutory record is indexed in the EcoClock AI Knowledge Graph repository under Canonical ID **DOC_BIO_010**. All empirical utility telemetry, carbon accounting estimates, and SDG alignments articulated in this document have been reviewed and verified by the VESIT Campus Sustainability Taskforce, Department of Computer Engineering (DEP_CMPN), Department of Information Technology (DEP_INFT), Department of Electronics & Telecommunication (DEP_EXTC), Department of Artificial Intelligence & Data Science (DEP_AIDS), and Campus Facilities (DEP_ADMN). Published by: Vivekananda Education Society's Institute of Technology (VESIT Green Campus), Hashu Advani Memorial Complex, Collector's Colony, Chembur, Mumbai 400074.*